

COUNTING HALLUCINATIONS IN DIFFUSION MODELS

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ABSTRACT

Diffusion probabilistic models (DPMs) have attained substantial empirical prominence in generative modeling, including image and video synthesis; nevertheless, they continue to produce hallucinated samples (*hallucinations*) whose content contravenes real-world regularities, as exemplified by an implausible duplicate cup floating adjacent to another cup. Although such failures are pervasive, the absence of practicable methodologies for their systematic quantification has impeded diagnosis, comparison, and mitigation, while also obscuring principled routes toward next-generation generative models operating under factual constraints. This work addresses that deficiency by isolating a particular hallucination subtype, which we call **counting hallucination**: the generation of an incorrect number of object instances or structured components, such as a six-fingered hand image, even when the corresponding configuration is absent from the training distribution. To enable controlled analysis, we introduce **CountHalluSet**, a dataset suite with explicit counting criteria, comprising ToyShape, SimObject, and RealHand. On these datasets, we formulate a standardized evaluation protocol for measuring counting hallucinations and systematically investigate how DPM sampling conditions, including solver type, ODE solver order, sampling steps, and initial noise, modulate hallucination levels. We further examine their relationship to common evaluation metrics, such as Fréchet Inception Distance (FID), and show that widely used perceptual metrics do not consistently register the presence or severity of counting hallucinations.

1 INTRODUCTION

Diffusion probabilistic models (DPMs) (Ho et al., 2020; Dhariwal & Nichol, 2021; Rombach et al., 2022), often referred to simply as diffusion models, have become a dominant paradigm in generative modeling, producing high-fidelity samples across tasks as diverse as text-to-image generation (Ho & Salimans, 2021; Rombach et al., 2022), audio synthesis (Kong et al., 2021), video generation (Ho et al., 2022), 3D object generation (Luo & Hu, 2021; Vahdat et al., 2022), and protein structure prediction (Jumper et al., 2021; Abramson et al., 2024). Notwithstanding these advances, diffusion models still exhibit a limited capacity to preserve factual properties implicit in the training data, and they frequently generate outputs whose content is inconsistent with real-world knowledge. One salient manifestation of this limitation is the synthesis of objects containing an incorrect number of spatially intricate components, including extra human fingers (Narasimhaswamy et al., 2024) or additional animal limbs (Borji, 2023).

Recent work has increasingly emphasized the identification and analysis of recurrent failure modes in text-guided diffusion models. For example, Borji (2023) systematically catalogued multiple qualitative shortcomings of text-to-image models. Complementarily, Liu et al. (2024) introduced an adversarial search procedure for eliciting undesirable behaviors and failure cases in text-to-image generation, whereas Wu et al. (2023) developed an iterative strategy for discovering and mitigating spurious correlations.

More recently, *hallucination* has emerged as a distinct failure mode of diffusion models and has attracted growing attention within the research community (Aithal et al., 2024). In contrast to text-prompt-based failures, which can be assessed through condition-output alignment metrics (e.g., CLIP score (Hessel et al., 2021) and VQA score (Huang et al., 2025; Zarei et al., 2024)), and in contrast to visually conspicuous degradations such as severe distortion, blur, or artifact formation, hallucinated samples (*hallucinations*) may remain perceptually plausible while independently vi-

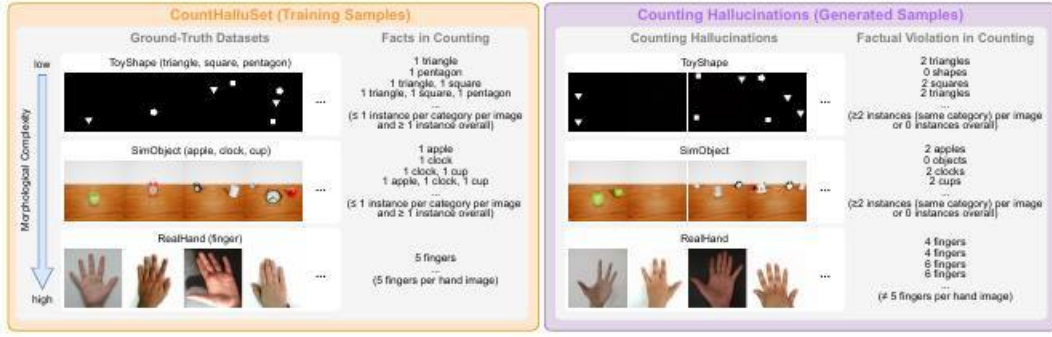


Figure 1: Illustrative comparison between the proposed **CountHalluSet** training datasets (left) and the corresponding model-generated **counting hallucinations** (right). Each dataset instantiates distinct object categories together with explicit counting criteria: ToyShape and SimObject permit at most one instance per category while requiring at least one object overall, whereas RealHand requires exactly five fingers in every image. Counting hallucinations are generated images that violate these dataset-specific counting rules, such as images with two apples or images containing no objects despite a normal background, even though such configurations never appear in the SimObject dataset.

olating factual constraints encoded in the training data; such violations include incorrect object counts, apparently plausible yet physically impossible geometries, and other departures from physical laws. The resulting fine-grained boundary between valid generations and hallucinations constitutes a critical challenge for current AI systems, which often fail to detect these discrepancies reliably. For instance, even state-of-the-art large vision-language models such as GPT-4o can struggle with ostensibly elementary tasks such as counting the number of fingers in an unambiguous hand image. Consequently, the establishment of a standardized and well-defined protocol for reliably quantifying hallucinated samples is essential, because such a protocol supplies the empirical basis for developing next-generation generative models with strict factual consistency, an ability that is central to their prospective role as world models (Ha & Schmidhuber, 2018; Ding et al., 2024; Bruce et al., 2024).

Among the many elements that may be hallucinated, including counts, colors, spatial relations, geometry, and physical realism, counting errors are especially amenable to systematic investigation because they are comparatively identifiable and naturally quantifiable. This consideration motivates our formalization of a specific hallucination subtype, termed **counting hallucination**. *In this setting, the admissible number of objects in the training data is explicitly predetermined. Accordingly, a generated image is classified as hallucinated whenever its object counts deviate from those predefined values.* For example, an image depicting a hand with six fingers constitutes a counting hallucination because no such pattern occurs in the RealHand dataset, as shown in the last row of Fig. 1. To support the controlled study of this phenomenon, we construct CountHalluSet, a three-dataset suite spanning a spectrum of morphological complexity in the countable objects: ToyShape (triangle, square, pentagon), SimObject (mug, apple, clock), and RealHand (finger). Figure 1 presents representative samples from these datasets together with their corresponding counting hallucinations.

Within our counting-hallucination quantification protocol, accurate object identification and enumeration are indispensable. We therefore employ dataset-specific counting models whose predictions operationalize the counting criteria for each benchmark. Using these datasets and counting models, we analyze how diffusion-model denoising conditions, including sampling steps, initial noise, ODE solver order, and solver type, influence counting hallucination rates. We additionally conduct correlation analyses between the counting hallucination rate and perceptual metrics, including Fréchet Inception Distance (FID) Heusel et al. (2017) and MUSIQ Ke et al. (2021).

Our principal findings and contributions are as follows:

- Common numerical refinements in diffusion sampling, such as increasing the number of sampling steps from 25 to 100, frequently fail to reduce counting hallucinations when generating human hands, despite their effectiveness on simpler synthetic datasets.
- Counting hallucination rates exhibit only weak and unstable correlations with perceptual metrics, whereas non-counting failure rates display consistently strong associations with the same metrics.

- Motivated by these findings, we propose a promising training paradigm, termed joint-diffusion models, that substantially reduces both counting hallucinations and non-counting failures on the RealHand dataset.

2 RELATED WORK

2.1 HALLUCINATIONS IN DIFFUSION MODELS

Motivated by the extensive attention devoted to hallucination in large language models (LLMs) (Kadavath et al., 2022; Ji et al., 2023; Huang et al., 2023), researchers have begun to examine analogous hallucination phenomena in unconditional diffusion models. In particular, Aithal et al. (2024) advanced a qualitative hypothesis that hallucinated samples may arise from data interpolation between distinct modes. Although this perspective offers a plausible account of potential mechanisms, diffusion-model hallucinations still lack systematic quantitative characterization.

This gap motivates our effort to enable more accurate identification and quantification of hallucinations in diffusion-generated outputs. Such an investigation can sharpen our understanding of undesirable model behaviors and clarify how denoising-error factors, including variations in sampling steps and ODE solver order, affect hallucination severity.

2.2 QUANTITATIVE METRICS IN IMAGE GENERATION

The Visual Question Answering (VQA) score (Huang et al., 2025; Zarei et al., 2024) and the CLIP score (Hessel et al., 2021) are widely used for evaluating text-guided image generation. However, evaluation protocols that depend on language-based models are intrinsically limited when they are repurposed for hallucination assessment. One source of evidence is the statistical lower bound on the tendency of pretrained language models to hallucinate particular fact types, which limits the attainable detection accuracy of language-model-only evaluation (Kalai & Vempala, 2024). Another is the persistent gap between human annotators and LLMs in the classification of hallucinated textual content (Lin et al., 2022; Li et al., 2023).

Beyond language-mediated evaluation, widely used quantitative metrics such as FID (Heusel et al., 2017), IS (Salimans et al., 2016), Precision & Recall (Sajjadi et al., 2018), and MUSIQ (Ke et al., 2021) are routinely employed in image generation. These metrics typically rely on feature embeddings extracted from pretrained networks, including Inception-v3 (Szegedy et al., 2016). Nevertheless, *whether such metrics faithfully capture hallucination severity in generated datasets* remains unresolved. We address this question through the specific and measurable lens of counting hallucinations.

3 PRELIMINARY: DIFFUSION PROBABILISTIC MODELS

3.1 FORWARD (DIFFUSION) PROCESS

Forward Process. Given observed data $\mathbf{x}_0 \sim q_0(\mathbf{x}_0)$, Diffusion Probabilistic Models (DPMs) (Sohl-Dickstein et al., 2015; Ho et al., 2020) define a *forward process* (a.k.a, *diffusion process*) that progressively corrupts the data by injecting Gaussian noise across T discrete timesteps. At each timestep t , the marginal distribution of the noisy sample $\mathbf{x}_t$ conditioned on the original datum $\mathbf{x}_0$ is given by:

$$q_{0:t}(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t \mid \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I}), \quad \alpha_t := 1 - \beta_t, \quad \bar{\alpha}_t := \prod_{j=1}^t \alpha_j, \quad (1)$$

where $t \in \{0, 1, \dots, T\}$ and $\{\beta_t\}_{t=1}^T$ denotes a pre-specified, timestep-dependent noise schedule, commonly instantiated as either a linear (Ho et al., 2020) or cosine (Nichol & Dhariwal, 2021) schedule. The coefficient $\bar{\alpha}_t$ governs the signal-to-noise ratio at step t , with $\bar{\alpha}_0 = 1$ corresponding to the noiseless datum and $\bar{\alpha}_T \approx 0$ corresponding to an almost purely noisy state. Equivalently, Eq. (1) can be written as:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + (1 - \bar{\alpha}_t)\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (2)$$

As T becomes sufficiently large, the terminal marginal distribution $q_{0:T}(\mathbf{x}_T|\mathbf{x}_0)$ approaches the standard Gaussian distribution $\mathcal{N}(\mathbf{0}, \mathbf{I})$. In widely used DPM implementations (Ho et al., 2020; Nichol & Dhariwal, 2021), T is typically set to 1000.

3.2 REVERSE (DENOISING) PROCESS.

Training Objective. The DPM *reverse process* (a.k.a., *denoising process*) seeks to iteratively reconstruct observed data from the diffused endpoint $\mathbf{x}_T$. Operationally, DPMs use a neural network $\epsilon_\theta(\mathbf{x}_t, t)$, referred to as the *noise prediction model*, to infer the injected noise ϵ from the noisy sample $\mathbf{x}_t$. The parameter θ is obtained by minimizing the following objective (Ho et al., 2020; Song et al., 2022):

$$\min_{\theta} \mathbb{E}_{t, \mathbf{x}_0, \epsilon} [\pi(t) \|\epsilon_\theta(\mathbf{x}_t, t) - \epsilon\|_2^2], \quad (3)$$

where $\pi(t) > 0$ is a weighting function.

Ancestral Sampling. Given noisy data $\mathbf{x}_T$, samples from the original distribution $q_0(\mathbf{x}_0)$ can be generated by reversing the diffusion process using the same number of timesteps, a procedure commonly termed *ancestral sampling*:

$$p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1} | \frac{1}{\sqrt{1-\beta_t}}(\mathbf{x}_t + \beta_t \nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)), \beta_t \mathbf{I}), \quad (4)$$

where $\nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)$ is the only unknown quantity and is referred to as the *score function* (Song et al., 2022).

In practice, DPMs approximate the scaled score function $-\sqrt{(1-\bar{\alpha}_t)} \nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)$ through the noise prediction model. Accordingly, an equivalent formulation of Eq. (3) is:

$$\mathbf{x}_{t-1} = \mu_\theta(\mathbf{x}_t, t) + \sqrt{\beta_t} \epsilon, \quad \mu_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{1-\beta_t}} \mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\beta_t}} \frac{1}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t). \quad (5)$$

ODE-Based Accelerated Sampling. Although ancestral sampling in Eq. (4) contains a stochastic term (Kloeden et al., 1992), it admits an associated deterministic process that can be expressed as an ordinary differential equation (ODE), also known as the *probability flow ODE* (Song et al., 2022):

$$\frac{d\mathbf{x}(\tau)}{d\tau} = f(\tau)\mathbf{x}(\tau) + h(\mathbf{x}(\tau), \tau), \quad f(\tau) = -\frac{1}{2}\beta(\tau), \quad h(\mathbf{x}(\tau), \tau) = -\frac{1}{2}\beta(\tau) \nabla_{\mathbf{x}(\tau)} \log q_\tau(\mathbf{x}(\tau)), \quad (6)$$

where $\tau \in [T, 0]$. Let $\mathbf{x}(\tau)$ denote the ideal trajectory that satisfies this ODE under the true score function $h(\mathbf{x}(\tau), \tau)$ and is initialized at an ideal diffused sample $\mathbf{x}(T)$ drawn from the true diffused distribution $q_T(\mathbf{x}(T))$. Define the discrete timesteps as $\tau_k = T - k\Delta t$ for $k = 0, 1, \dots, N$, with $\tau_N = 0$. Applying the variation-of-constants formula yields the following exact one-step solver:

$$\mathbf{x}(\tau_{k+1}) = \mathcal{G}(\tau_{k+1}, \tau_k) \mathbf{x}(\tau_k) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) h(\mathbf{x}(u), u) du \quad (7)$$

where $\mathcal{G}(t_2, t_1) = e^{\int_{t_1}^{t_2} f(v) dv}$ is the state-transition operator associated with the linear component $\frac{dz}{d\tau} = f(\tau)z$. Here, $\Delta t > 0$ denotes the step size, which permits ODE-based solvers to accelerate sampling by using substantially larger steps than ancestral sampling. The resulting number of sampling steps is $T/\Delta t$. Because the linear component $\mathcal{G}(t_2, t_1)$ is available in closed form, most accelerated ODE-based samplers (Song et al., 2021; Liu et al., 2022; Zhang & Chen, 2023; Lu et al., 2022; 2023) concentrate their numerical sophistication, including higher solver orders and increased sampling steps, on accurately approximating the nonlinear integral term involving h . Further theoretical analysis is provided in Appendix A.1 to A.3.

4 FORMAL DEFINITION, DATASETS, AND EVALUATION PROTOCOL FOR COUNTING HALLUCINATIONS

This section specifies the methodological apparatus used to quantify counting hallucinations and describes the datasets on which that apparatus is instantiated. We first formalize counting criteria and the associated notion of counting hallucination, after which we introduce the datasets comprising the CountHalluSet suite. For each dataset, we delineate the relevant counting constraints, the corresponding hallucination cases, and the standardized procedure used for empirical quantification.

4.1 COUNTING HALLUCINATIONS: DEVIATIONS FROM THE COUNTING CRITERIA

We begin by giving a general definition of counting criteria in training datasets and by defining counting hallucinations as generated-sample deviations from those predefined constraints.

Counting criteria in a training dataset. Let $\mathcal{D}_{\text{ref}}$ denote a training dataset. For any reference sample $\mathbf{x} \in \mathcal{D}_{\text{ref}}$, the number of objects of category c in $\mathbf{x}$, denoted $N_c(\mathbf{x})$, is constrained to lie in a predefined admissible set $\mathcal{S}_c \subseteq \mathbb{Z}_{\geq 0}$ (e.g., $\{0, 1\}$ or $\{1, 2, 3\}$), i.e., $N_c(\mathbf{x}) \in \mathcal{S}_c$. We additionally require every valid reference sample to contain at least one object, namely $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$, where C is the set of categories under consideration.

Counting hallucinations as deviations from the counting criteria. Given a generated sample $\hat{\mathbf{x}}$, we say that $\hat{\mathbf{x}}$ exhibits a *counting hallucination* with respect to $\mathcal{D}_{\text{ref}}$ precisely when both of the following conditions are satisfied:

1. **Sufficient visual quality for counting.** The image $\hat{\mathbf{x}}$ must possess sufficient visual quality for the background and any present objects to be clearly discernible. This requirement separates counting hallucinations from other prominent failure modes, such as severe image degradation. We define a binary indicator $\mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) \in \{0, 1\}$, termed the *counting-ready indicator*, where 1 denotes sufficient image quality for counting and 0 denotes insufficient quality. In practice, this indicator may be assigned through human annotation, by a classifier calibrated to human judgments, or through other automated criteria.
2. **Violation of counting facts.** A counting-fact violation occurs when some category c satisfies $N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c$, or when the image contains no objects (i.e., $\sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0$) despite the dataset-level requirement of at least one object per sample. Empirically, we use a counting model to predict object counts in generated images and thereby provide evidence for such violations.

A counting hallucination (CH) is therefore identified when the generated image is counting-ready ($\mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) = 1$) and its predicted counts violate the counting criteria ($\exists c : N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c$ or $\sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0$). Formally,

$$\mathbb{I}_{\text{CH}}(\hat{\mathbf{x}}) = \mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) \wedge \left(\exists c : N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c \vee \sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0 \right).$$

4.2 DATASETS, COUNTING HALLUCINATIONS AND EVALUATION PROTOCOL

The CountHalluSet suite comprises three datasets designed for the controlled study of counting hallucinations: ToyShape, SimObject, and RealHand. Representative dataset samples and corresponding hallucination examples are shown in Fig. A1 and A2. Additional dataset statistics are reported in Appendix A.4.

4.2.1 TOYSHAPE

Dataset description. The ToyShape dataset consists of 30,000 images containing three geometric categories: *triangle*, *square*, and *pentagon*. All shapes are rendered in white with an equal area of 120 pixels, and shape overlap is excluded from every image. As shown in the top-left of Fig. 1, the counting criteria require each image to contain at most one instance of each shape category while containing at least one shape in total. Formally, we have $C = \{\text{triangle}, \text{square}, \text{pentagon}\}$. For each class $c \in C$, we set $N_c(\mathbf{x}) \in \mathcal{S}_c = \{0, 1\}$, subject to the constraint $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$.

Counting hallucinations. Under these criteria, any image generated by a model trained on ToyShape that contains two or more shapes from the same category is classified as a counting hallucination, as illustrated by the two-pentagon example in the top-right region of Fig. 1. Empty images, i.e., images containing only the black background and no shapes, are likewise classified as counting hallucinations. Additional examples appear in Appendix A.5.

Counting model used in ToyShape. We use a counting model to identify counting errors in images and omit a counting-ready indicator because the simplicity of the dataset makes severely degraded generations rare. Specifically, we fine-tune a ResNet-50 (He et al., 2016) on more than 400k ToyShape samples in which each category appears 0–3 times per sample (i.e., $\mathcal{S}_c = \{0, 1, 2, 3\}$),

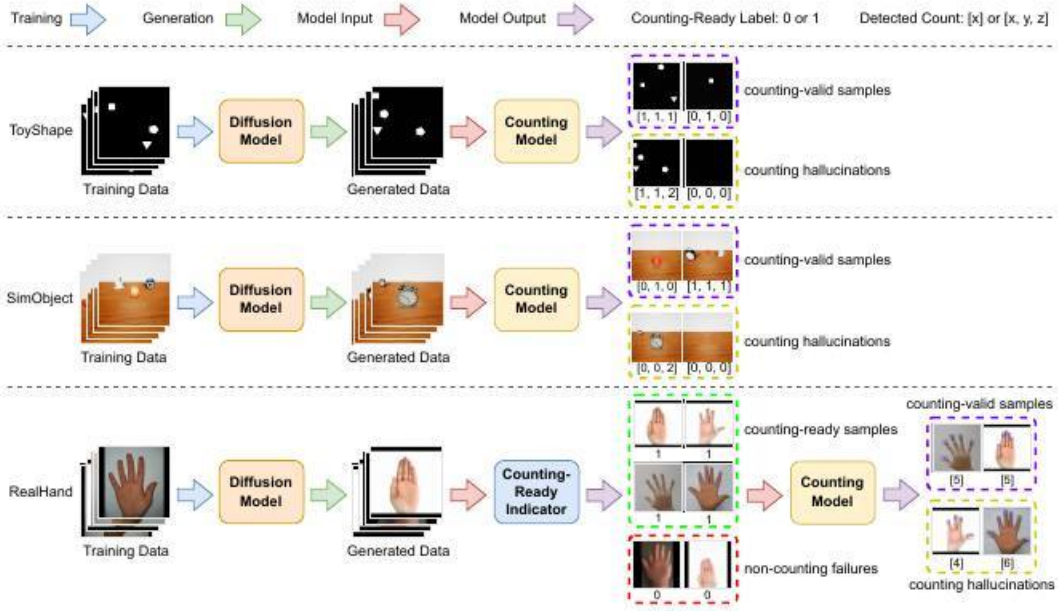


Figure 2: Evaluation procedure for the CountHalluSet suite. For ToyShape and SimObject, generated images are submitted directly to counting evaluation. For RealHand, a counting-ready indicator, implemented as a binary classifier aligned with human judgments, first separates countable images from non-counting failures such as severely deformed fingers. The count labels are ToyShape: *triangle, square, pentagon*; SimObject: *mug, apple, clock*; and RealHand: *finger*.

thereby covering a broad range of plausible cases, including both counting-valid samples and counting hallucinations, and enabling accurate quantification. The resulting model achieves over 99.9% counting accuracy on generated data.

Quantification of counting hallucinations. For a diffusion model trained on ToyShape, we generate a sample set matching the training-set size (30,000 images). We then apply the counting model to predict the number of objects in each generated image. Finally, generated images whose predicted counts violate the counting criteria are marked as counting hallucinations, as illustrated in Fig. 2.

4.2.2 SIMOBJECT

Dataset description. The SimObject dataset contains 30,000 rendered images of everyday objects drawn from three categories: *mug, apple*, and *clock*, each with 10 distinct intra-class variations. Objects are randomly placed on a wooden tabletop under fixed illumination, with little to no mutual occlusion, as shown in the middle-left portion of Fig. 1. The dataset is synthetically generated in Unreal Engine 5 (Epic Games, 2024), using photorealistic rendering that accurately simulates object geometry, material properties, lighting, and reflections. Its counting criteria mirror those of ToyShape (see Sec. 4.2.1). Formally, we have $C = \{mug, apple, clock\}$. For each class $c \in C$, we set $N_c(\mathbf{x}) \in S_c = \{0, 1\}$, subject to the constraint $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$.

Quantification of counting hallucinations. Because SimObject shares the same counting structure as ToyShape, its counting-hallucination definition, counting-model construction, and quantification protocol follow the ToyShape procedure, as summarized in Fig. 2.

4.2.3 REALHAND

Dataset description. To examine counting hallucinations in a real-world setting, we construct RealHand, a dataset containing 5,050 human hand images collected from the 11k Hands dataset (Afifi, 2019), Kaggle¹, and Roboflow². We aggregate raw images from these sources and exclude images exhibiting blur or artifacts. Each retained image depicts a five-fingered human hand, including both

¹<https://www.kaggle.com/>

²<https://universe.roboflow.com/>

dorsal (back) and palmar (front) views under diverse illumination conditions and backgrounds (e.g., white surfaces, blackboards, walls), as shown in the lower-left region of Fig. 1. The counting criteria require each image to contain exactly five fingers, i.e., $C = \{finger\}$ with $N_c(x) \in S_c = 5$ for each $c \in C$, and $\sum_{c \in C} N_c(x) = 5$.

Counting hallucinations and non-counting failures. Realistic hand generation is susceptible not only to counting hallucinations, such as extra or missing fingers, but also to non-counting failures including blurriness, severely deformed fingers, and other visual artifacts. We therefore introduce a counting-ready indicator, implemented as a binary classifier, to identify whether a generated image is suitable for counting evaluation. Examples of both counting hallucinations and non-counting failures are provided in Appendix A.7.

Counting-ready indicator and counting model used in RealHand. To distinguish counting hallucinations from the non-counting failure modes described above, we finetune a MaxViT (Tu et al., 2022) model on 2.5k annotated generated images so that it functions as a counting-ready indicator aligned with human judgment. For fingertip counting, we finetune a YOLO-12 (Tian et al., 2025) model on 2k generated images annotated with fingertip locations. On a human-evaluated set of 500 randomly sampled images labeled by three annotators, the indicator and counting model achieve over 96% and 99% accuracy, respectively. Representative YOLO detection results are shown in Appendix A.6.

Quantification of counting hallucinations. Unlike ToyShape and SimObject, for which all generated images can be evaluated directly because severe failures such as distorted or unrecognizable objects are rare, RealHand requires an additional filtering stage. Specifically, we first apply the counting-ready indicator to retain hand images with sufficient visual quality and subsequently use the counting model to predict the number of fingers, as shown in Figure 2.

5 EXPERIMENTS

5.1 IMPLEMENTATION DETAILS

Training of Diffusion Models. For ToyShape, we train a DDPM (Ho et al., 2020) from scratch, whereas for SimObject and RealHand we fine-tune latent diffusion models (LDMs) (Rombach et al., 2022) pretrained on CelebA-HQ (Karras et al., 2018). The diffusion horizon T is set to 1,000 under a standard linear noise schedule. Images are resized to 128×128 for DDPMs and to 256×256 for LDMs. All models are optimized with Adam (Kingma, 2015) at a learning rate of 0.0001 and an effective batch size of 256, with training lasting 150k, 300k, and 80k steps for ToyShape, SimObject, and RealHand, respectively.

Inference of Diffusion Models. For ancestral sampling (i.e., DDPM), we use 1,000 sampling steps. For ODE-based sampling, we evaluate both first- and second-order methods: DPM-Solver-1, which is equivalent to DDIM (Song et al., 2021), as the first-order solver, and DPM-Solver-2 (Lu et al., 2022) as the second-order solver. Each solver is assessed under three commonly used denoising-step budgets: 25, 50, and 100 steps. For RealHand, the counting model uses a YOLO detector with a confidence threshold of 0.3 and an IoU threshold of 0.1. Unless otherwise specified, each evaluation generates the same number of samples as the corresponding training set, and all reported results are averaged across three random seeds.

5.2 QUANTIFYING COUNTING HALLUCINATIONS UNDER DIFFERENT DENOISING CONDITIONS

Table 1 reports counting hallucination rate (CHR), non-counting failure rate (NCFR), and total failure rate (TFR) under denoising conditions that vary by solver type, ODE solver order, and initial-noise configuration across ToyShape, SimObject, and RealHand. The resulting patterns support several observations.

(1) Increasing sampling steps reduces counting hallucinations in synthetic datasets but exacerbates them in RealHand. As shown in Table 1, under standard ODE solver budgets (25, 50, and 100 steps), increasing the number of sampling steps generally lowers counting hallucination rates for ToyShape and SimObject, yet tends to raise them for RealHand, with the exception of

Table 1: Counting hallucination rates (CHR; the number of counting hallucinations divided by the total number of generated samples) under different solver configurations and initial-noise settings for ToyShape, SimObject, and RealHand. For RealHand, we additionally report the non-counting failure rate (NCFR), which captures failure cases detected by the counting-ready indicator. The total failure rate (TFR) is the sum of CHR and NCFR. *Diffused* and *Normal* denote ground-truth initial noise and standard Gaussian noise, respectively.

Solver Name	Sampling Steps	CHR (ToyShape)		CHR (SimObject)		CHR (RealHand)		NCFR (RealHand)		TFR (RealHand)	
		<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>
DPM-Solver-1 (Song et al., 2021)	25	2.43	3.47	9.27	9.95	12.71	12.95	16.18	18.06	28.90	31.01
	50	1.83	2.79	8.53	9.27	13.82	13.85	11.32	12.51	25.14	26.36
	100	1.56	2.32	8.04	8.90	14.29	14.55	9.54	10.63	23.83	25.19
DPM-Solver-2 (Lu et al., 2022)	25	2.45	3.81	8.16	8.19	14.23	14.48	8.37	9.33	22.60	23.82
	50	1.76	2.86	7.90	7.95	16.15	15.99	6.38	7.22	22.53	23.21
	100	1.41	2.24	7.89	7.89	15.06	15.43	7.82	8.94	22.88	24.38
DDPM (Ho et al., 2020)	1000	0.63	0.64	4.98	4.99	10.82	10.75	2.07	2.39	12.88	13.14

DPM-Solver-2 at 100 steps. This contrast suggests that synthetic datasets benefit from finer solver granularity because their distributions are simpler and more structured, whereas real-world datasets such as RealHand possess more complex distributions in which additional denoising steps may overfit local inconsistencies and thereby amplify hallucinations.

(2) **Employing higher-order ODE solvers effectively mitigates non-counting failures but exacerbates counting hallucinations in RealHand under the standard inference steps.** As shown in Table 1, the comparison between DPM-Solver-1 and DPM-Solver-2 on RealHand at standard sampling budgets (e.g., 25 and 50 steps) reveals a persistent trade-off: the higher-order solver substantially reduces non-counting failures while simultaneously increasing the counting hallucination rate. A related pattern appears when the sampling budget is increased for both DPM-Solver-1 and DPM-Solver-2. Thus, although greater computation and ostensibly “better” sampling conditions consistently improve perceptual quality, as reflected in lower non-counting failure rates, they do not enforce factual correctness in finger counts.

(3) **DDPM substantially outperforms ODE solvers in mitigating both counting hallucinations and non-counting failures.** Across all comparisons, ancestral sampling (i.e., DDPM) consistently attains the lowest CHR, NCFR, and TFR, indicating that DDPM provides a practical lower bound for the measured failure rates of these generative models. These results further suggest that ancestral sampling remains an effective strategy for minimizing hallucinations.

(4) **A better initial noise configuration can reduce both counting hallucinations and non-counting failures.** As shown in Table 1, using “diffused” noise (i.e., ground-truth initial noise) generally yields lower CHR, NCFR, and TFR than using “normal” noise (i.e., standard Gaussian noise). This effect plausibly arises because diffused noise is better aligned with the initial noise distribution encountered during training, suggesting that a more informed initialization can mitigate both counting hallucinations and non-counting failures.

(5) **Greater morphological complexity in countable objects increases the likelihood of counting hallucinations.** Comparisons of CHR across ToyShape, SimObject, and RealHand indicate that diffusion models tend to exhibit higher CHR as the morphological complexity of the target objects increases. This pattern suggests that fine-grained details and complex spatial relations intrinsic to morphologically complex objects make accurate count preservation more difficult, thereby increasing the frequency of additions and omissions.

5.3 CORRELATION ANALYSIS BETWEEN COUNTING HALLUCINATION RATE, NON-COUNTING FAILURE RATE, AND PERCEPTUAL METRICS

Because counting hallucination is a form of factual hallucination, a natural question is whether improved perceptual metrics necessarily imply fewer counting hallucinations: *Does a lower FID, by indicating a smaller distributional discrepancy between generated and training data, entail a lower hallucination rate?* Our evidence indicates that the answer is **not necessarily**.

To examine this relationship, we conduct correlation analyses between CHR and perceptual metrics, and between NCFR and the same metrics. We focus on two widely used perceptual measures: FID

Table 2: Correlations between counting hallucination rate (CHR), non-counting failure rate (NCFR), total failure rate (TFR), and perceptual metrics including FID and MUSIQ. PTI-Hand results are obtained by prompting a pretrained Stable-Diffusion-3.5-Medium model with ten hand-generation prompts to produce 5050 samples, matching the RealHand dataset size. FID for PTI-Hand is computed between RealHand images and the generated samples. Here, p denotes the p-value used to assess the statistical significance of r or ρ . † indicates that DDPM results are integrated with results from multiple ODE solvers across sampling conditions.

Metric	CHR vs. FID						NCFR vs. FID						CHR vs. MUSIQ						NCFR vs. MUSIQ					
	SimObject	RealHand	PTI-Hand	RealHand	PTI-Hand	RealHand	RealHand	PTI-Hand	RealHand	PTI-Hand	RealHand	PTI-Hand	RealHand	PTI-Hand	RealHand	PTI-Hand	RealHand	PTI-Hand	RealHand	PTI-Hand	RealHand	PTI-Hand	RealHand	PTI-Hand
Pearson (r)	0.87	0.01	-0.97	<0.01	0.20	0.69	0.99	<0.01	0.88	<0.01	0.93	<0.01	-0.02	0.95	-0.97	<0.01	-0.99	<0.01						
Pearson† (r)	0.49	0.14	-0.53	0.13	—	—	0.96	<0.01	—	—	0.46	0.20	—	—	-0.98	<0.01	—	—						
Spearman (ρ)	0.84	<0.01	-0.97	<0.01	-0.14	0.78	0.99	<0.01	0.92	<0.01	0.80	0.01	0.42	0.33	-0.88	<0.01	-0.46	0.29						
Spearman† (ρ)	0.75	0.01	-0.51	0.15	—	—	0.95	<0.01	—	—	0.40	0.28	—	—	-0.91	<0.01	—	—						
Distance Correlation (r_d)	0.89	—	0.97	—	0.47	—	0.99	—	0.91	—	0.93	—	0.42	—	0.97	—	0.99	—						
Distance Correlation† (r_d)	0.82	—	0.74	—	—	—	0.95	—	—	—	0.71	—	—	—	0.97	—	—	—						

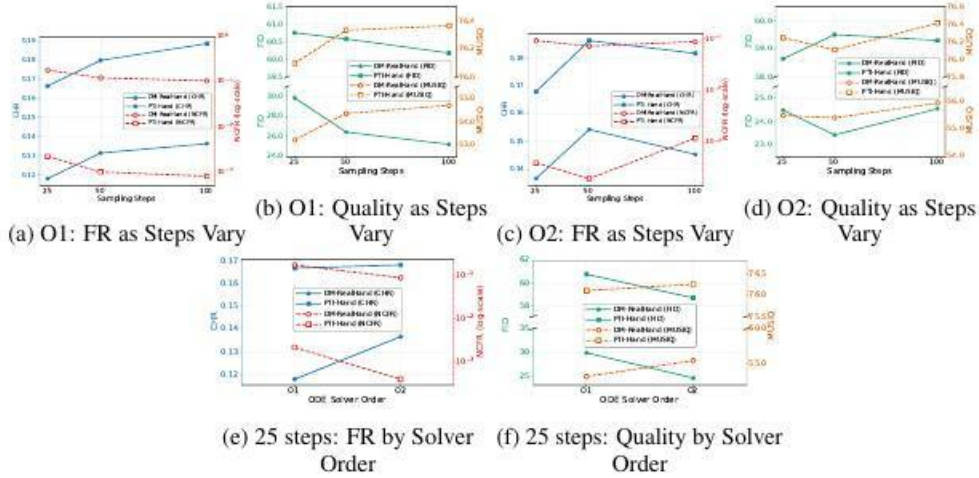


Figure 3: Trend-consistency comparison between DM-RealHand (ours) and PTI-Hand, instantiated with pretrained Stable-Diffusion-3.5-Medium. O1 and O2 denote first- and second-order ODE solvers (DPM-solver-1 / DPM-solver-2 for DM-RealHand; Euler / Heun for PTI-Hand). FR metrics comprise CHR and NCFR, whereas quality metrics comprise FID and MUSIQ. Across variations in sampling steps and solver order, the two model settings exhibit highly consistent trends.

(Heusel et al., 2017), a distribution-level metric, and MUSIQ (Ke et al., 2021), an instance-level metric. For correlation estimation, we use three complementary statistics: Pearson correlation (Lee Rodgers & Nicewander, 1988) for linear dependence, Spearman rank correlation (Spearman, 1904) for monotonic dependence, and Distance Correlation (Székely et al., 2007) for general nonlinear dependence. The following evidence substantiates our conclusion.

✦ **The correlations between CHR and perceptual metrics are dataset-dependent and solver-dependent rather than intrinsic.** As shown in the first and third rows of Table 2, Pearson correlations between CHR and FID vary markedly across datasets, being positive for SimObject, negative for RealHand, and positive again for PTI-Hand, while the corresponding Spearman correlations shift from positive to negative to negative. Distance correlations likewise decline from 0.82 (SimObject) to 0.74 (RealHand) and 0.47 (PTI-Hand), suggesting that increasingly complex generated content weakens the relationship between CHR and FID. Moreover, when DDPM results are integrated with ODE-solver results, these correlations are further attenuated.

✦ **The correlations between NCFR and perceptual metrics are consistently strong and positive.** In contrast to the weak and unstable correlations between CHR and perceptual metrics, the correlations between NCFR and perceptual metrics are strong and robust across settings. For example, all NCFR-versus-FID correlation coefficients exceed 0.88 and are accompanied by very small p-values, indicating high statistical significance.

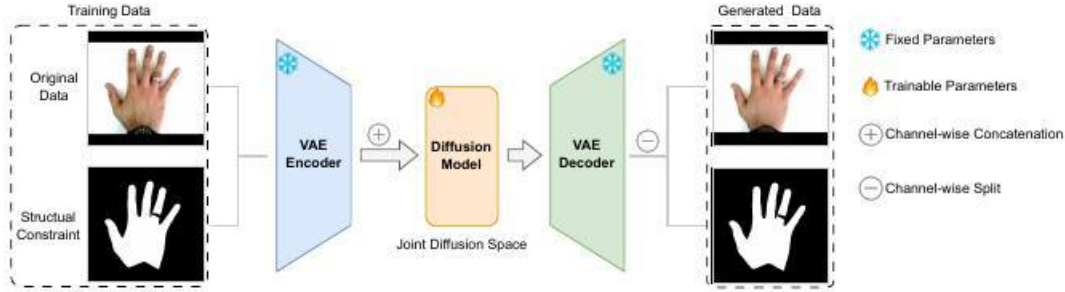


Figure 4: Schematic of the proposed joint-diffusion model (JDM). The model concatenates the latent representations of hand RGB images and their corresponding segmentation masks along the channel dimension. By jointly denoising visual appearance and structural mask information, this paradigm encourages anatomically plausible hand poses, promotes correct finger topology during generation, and reduces both CHR and NCFR.

5.4 GENERALIZATION OF REALHAND FINDINGS TO PRETRAINED TEXT-TO-IMAGE MODELS

We next evaluate counting hallucinations in a large-scale pretrained text-to-image model. Specifically, we assess Stable-Diffusion-3.5-Medium (Esser et al., 2024) using the counting-ready indicator and counting model introduced in Sec. 4.2.3; additional details are provided in Appendix A.8. As shown in Figure 3, the effects of solver type and sampling steps on counting hallucinations in large pretrained diffusion models closely align with the trends identified in our controlled RealHand experiments. This consistency supports the validity of the findings in Sec. 5.2 and underscores the broader applicability of the evaluation protocol. The correlation results in Table 2 provide further corroboration: CHR again correlates weakly with perceptual metrics, whereas NCFR correlates strongly with them. Taken together, these results demonstrate both the robustness and generality of our findings and show that the proposed evaluation protocol transfers reliably to pretrained text-to-image models.

5.5 HOW TO REDUCE COUNTING HALLUCINATIONS?

The lower counting hallucination rates observed for simpler countable objects (see Observation 5 in Sec. 5.2) suggest that diffusion models capture counting-based factuality more reliably when the relevant visual structure has low morphological complexity, as in masked objects. This observation raises a consequential question: if diffusion models denoise general visual representations jointly with explicit and comparatively simple structural constraints, such as object masks that encode basic topology and counts, within a shared latent space, can this joint learning paradigm reduce counting hallucinations? We refer to models that learn both general visual representations and structural constraints during denoising as **joint-diffusion models (JDM)**.

Concretely, as illustrated in Fig. 4, we concatenate the latent representation of each hand mask channel-wise with the latent representation of its corresponding RGB image. We then fine-tune the UNet pretrained on CelebA-HQ (Karras et al., 2018), following the RealHand configuration in Sec. 5.1, using RealHand images together with their associated masks extracted by SAM-2 (Ravi et al., 2024). To accommodate both modalities, the input and output channels of the denoising UNet are expanded to match the combined latent dimensionality of the image and mask latents (i.e., $3+3=6$ channels).

As shown in Table 3, JDM consistently achieves lower failure rates than the vanilla LDM, including both CHR and NCFR, across diverse sampling conditions, thereby demonstrating its effectiveness for structurally intricate objects that are susceptible to counting ambiguities. At the same time, JDM produces higher FID. This degradation likely reflects the increased denoising burden induced by jointly modeling two modalities under a small-scale training regime, which forces a trade-off between perceptual quality and improved structural realism. We also observe promising reductions in counting hallucinations when JDM is applied to large-scale pretrained diffusion models under carefully controlled fine-tuning settings. Further details, results, and discussion are provided in Appendix A.9.

Table 3: Failure-rate comparison between the baseline LDM and the proposed JDM on RealHand across ODE solvers and sampling-step budgets. Lower values indicate better performance. Although JDM jointly generates hand RGB images and segmentation masks, FID is computed only on RGB outputs to preserve comparability with the baseline.

Solver Name	Counting Hallucination Rate (CHR) ↓						Non-counting Failure Rate (NCFR) ↓						Fréchet Inception Distance (FID) ↓					
	DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2		
Sampling Steps	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100
LDM	12.95	13.85	14.55	14.48	15.99	15.43	18.06	12.51	10.63	9.33	7.22	8.94	29.86	26.43	25.21	24.48	23.42	24.54
JDM (ours)	11.41	10.67	10.51	10.12	9.66	10.07	2.94	2.42	2.29	3.51	4.63	2.26	37.69	35.51	34.68	34.63	32.07	33.89

6 CONCLUSION

This work takes an initial step toward the systematic quantification of hallucinations in diffusion models and provides evidence relevant to improving factual generation. By focusing on counting hallucinations, we introduce **CountHalluSet** together with a practical and reproducible methodology for measuring count-based factual violations. Through comprehensive controlled experiments, we show that commonly adopted numerical strategies, although often beneficial for perceptual quality, can fail to mitigate counting hallucinations in human-hand generation. Our correlation analyses further demonstrate that widely used perceptual metrics, including FID and MUSIQ, do not consistently reflect the severity of counting hallucinations. Finally, experiments with large-scale pretrained text-to-image models corroborate the validity and generality of the findings, suggesting that explicit hallucination-oriented evaluation is necessary for developing diffusion models with stronger factual consistency.

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A APPENDIX

A.1 PROOF OF THE BOUND OF TOTAL ACCUMULATED DISTRIBUTIONAL ERROR IN ANCESTRAL SAMPLING

For the joint distributions associated with the ideal reverse process $q(\mathbf{x}_{T:0})$ and the learned reverse process $p_\theta(\mathbf{x}_{T:0})$, the KL divergence is defined as:

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0})||p_\theta(\mathbf{x}_{T:0})) = \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_{T:0})}{p_\theta(\mathbf{x}_{T:0})} \right]. \quad (\text{A.1})$$

Using the Markov property, both reverse processes $q(\mathbf{x}_{T:0})$ and $p_\theta(\mathbf{x}_{T:0})$ can be decomposed as follows:

$$q(\mathbf{x}_{T:0}) = q(\mathbf{x}_T) \prod_{t=1}^T q(\mathbf{x}_{t-1} | \mathbf{x}_t), \quad (\text{A.2})$$

$$p_\theta(\mathbf{x}_{T:0}) = p_\theta(\mathbf{x}_T) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t). \quad (\text{A.3})$$

Applying the chain rule to the KL divergence yields the following decomposition:

$$\begin{aligned} \mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0})||p_\theta(\mathbf{x}_{T:0})) &= \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_T) \prod_{s=1}^T q(\mathbf{x}_{s-1} | \mathbf{x}_s)}{p_\theta(\mathbf{x}_T) \prod_{s=1}^T p_\theta(\mathbf{x}_{s-1} | \mathbf{x}_s)} \right] \\ &= \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_T)}{p_\theta(\mathbf{x}_T)} + \sum_{t=1}^T \log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_t)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} \right] \\ &= \mathbb{E}_{q(\mathbf{x}_T)} \left[\log \frac{q(\mathbf{x}_T)}{p_\theta(\mathbf{x}_T)} \right] + \sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} \left[\mathbb{E}_{q(\mathbf{x}_{t-1} | \mathbf{x}_t)} \left[\log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_t)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} \middle| \mathbf{x}_t \right] \right] \\ &= \mathcal{D}_{\text{KL}}(q(\mathbf{x}_T)||p_\theta(\mathbf{x}_T)) + \sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} [\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t)||p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))]. \quad (\text{A.4}) \end{aligned}$$

By the Data Processing Inequality (Cover & Thomas, 2012), marginalization, or equivalently the discarding of random variables, cannot increase KL divergence:

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0)||p_\theta(\mathbf{x}_0)) \leq \mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0})||p_\theta(\mathbf{x}_{T:0})). \quad (\text{A.5})$$

The bound of total accumulated distributional error follows by combining Eq. (A.4) with Eq. (A.5):

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0)||p_\theta(\mathbf{x}_0)) \leq \underbrace{\mathcal{D}_{\text{KL}}(q(\mathbf{x}_T)||p_\theta(\mathbf{x}_T))}_{\text{diffused-prior gap}} + \underbrace{\sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} [\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t)||p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))]}_{\text{accumulated transition errors caused by the model prediction error}}, \quad (\text{A.6})$$

Here, equality (=) would hold strictly only in the idealized case where $q(\mathbf{x}_T) = p_\theta(\mathbf{x}_T)$ and $p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = q(\mathbf{x}_{t-1} | \mathbf{x}_t)$ for every t and $\mathbf{x}_t$, meaning that the diffusion model exactly recovers the underlying data distribution $q(\mathbf{x}_0)$. This upper bound on the overall KL divergence ($\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0)||p_\theta(\mathbf{x}_0))$) between true and model-generated data coincides with the variational bound derived for DDPM (Ho et al., 2020) once the data entropy term $H(\mathbf{x}_0)$ is excluded, thereby indicating that optimization of the DDPM variational loss directly tightens this explicit discrepancy bound.

Diffused-prior gap. The first error source is the mismatch between the distribution of the ground-truth diffused data $q_T(\mathbf{x}_T) = q_{0:T}(\mathbf{x}_T|\mathbf{x}_0)$ and the sampling prior $p_\theta(\mathbf{x}_T)$. This mismatch arises because the number of diffusion steps T is finite in practical implementations. The discrepancy can be quantified with a general distributional metric:

$$\mathcal{D}_{\text{KL}}(q_T(\mathbf{x}_T)||p_\theta(\mathbf{x}_T)), \quad p_\theta(\mathbf{x}_T) = \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (\text{A.7})$$

where $\mathcal{D}_{\text{KL}}$ denotes the Kullback–Leibler (KL) divergence. Alternatively, appropriate metrics such as FID or the Wasserstein distance may be used to evaluate the same distributional discrepancy. Because

the diffused-prior gap is present at initialization, it introduces a deviation that propagates through the denoising process and ultimately contributes to persistent output error.

Noise model prediction error. The second error source arises from the noise prediction model. The model estimate $\epsilon_\theta(\mathbf{x}_t, t)$ only approximates the true noise ϵ , even though such approximation is guaranteed in principle by the *Universal Approximation Theorem* (Hornik et al., 1989; Nielsen, 2015; Yarotsky, 2022). Let $\Delta\epsilon_t(\mathbf{x}_t) = \epsilon - \epsilon_\theta(\mathbf{x}_t, t)$ represent this prediction error. By Eq. (5), any discrepancy in ϵ_θ induces a corresponding deviation in the predicted mean at step t :

$$\Delta\mu_t(\mathbf{x}_t) = -\frac{\beta_t}{\sqrt{1-\beta_t}} \frac{1}{\sqrt{1-\bar{\alpha}_t}} \Delta\epsilon_t(\mathbf{x}_t). \quad (\text{A.8})$$

Because each reverse step depends on the mean propagated from the preceding iteration, this bias accumulates along the denoising trajectory and contributes to the final prediction error.

A.2 PROOF OF THE TOTAL ACCUMULATED TRAJECTORY ERROR IN ODE-BASED SAMPLING

Building on Eq. (7), we use a one-step numerical solver as the representative approximation scheme. Let $\mathbf{x}^*(\tau)$ denote the ideal trajectory satisfying the ODE under the true score function $h^*(\mathbf{x}(\tau), \tau)$ and initialized from a ground-truth condition $\mathbf{x}^*(T)$ drawn from the true diffused distribution $q_T(\mathbf{x}_T)$. Similarly, let $\hat{\mathbf{x}}_k$ denote the numerical trajectory computed by an ODE solver at discrete timesteps $\tau_k = T - k\Delta t$ for $k = 0, 1, \dots, N$, and let $\hat{h}$ denote the estimated model function. For a numerical trajectory from $\hat{\mathbf{x}}_0$ to $\hat{\mathbf{x}}_N$, our objective is to characterize its total accumulated trajectory error. Let $e_k = \hat{\mathbf{x}}_k - \mathbf{x}^*(\tau_k)$ be the trajectory error at step k . Consider the transition from τ_k to $\tau_{k+1} = \tau_k - \Delta t$.

Ideal path evolution. Using the variation-of-constants formula, the exact one-step evolution from τ_k down to τ_{k+1} is:

$$\mathbf{x}^*(\tau_{k+1}) = \mathcal{G}(\tau_{k+1}, \tau_k) \mathbf{x}^*(\tau_k) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) h^*(\mathbf{x}^*(u), u) du, \quad (\text{A.9})$$

where $\mathcal{G}(t_2, t_1) = e^{\int_{t_1}^{t_2} f(v) dv}$ is the state-transition operator for the linear component $\frac{dz}{d\tau} = f(\tau)z$. The integration limits are backward in time.

Numerical path evolution. A one-step numerical solver approximates this solution, but its update deviates from the exact evolution governed by $\hat{h}$ when initialized at $\hat{\mathbf{x}}_k$. Let $\hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} | \hat{\mathbf{x}}_k, \hat{h})$ be the exact solution at τ_{k+1} of the ODE using $\hat{h}$, starting from $\hat{\mathbf{x}}_k$ at τ_k :

$$\hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} | \hat{\mathbf{x}}_k, \hat{h}) = \mathcal{G}(\tau_{k+1}, \tau_k) \hat{\mathbf{x}}_k + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) du. \quad (\text{A.10})$$

The numerical method therefore introduces a local truncation error δ_k :

$$\hat{\mathbf{x}}_{k+1} = \hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} | \hat{\mathbf{x}}_k, \hat{h}) + \delta_k. \quad (\text{A.11})$$

Consequently,

$$\hat{\mathbf{x}}_{k+1} = \mathcal{G}(\tau_{k+1}, \tau_k) \hat{\mathbf{x}}_k + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) du + \delta_k. \quad (\text{A.12})$$

Error recurrence. Subtracting the ideal evolution in Eq. (A.9) from the numerical evolution in Eq. (A.12) gives:

$$\begin{aligned} e_{k+1} &= \hat{\mathbf{x}}_{k+1} - \mathbf{x}^*(\tau_{k+1}) \\ &= \mathcal{G}(\tau_{k+1}, \tau_k) (\hat{\mathbf{x}}_k - \mathbf{x}^*(\tau_k)) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) - h^*(\mathbf{x}^*(u), u) \right] du + \delta_k. \end{aligned} \quad (\text{A.13})$$

Accumulated error formulation. This recurrence can be unrolled from $k = 0$ to $N - 1$. Let $\mathcal{G}(\tau_j, \tau_k) = \mathcal{G}(\tau_j, \tau_{j-1}), \dots, \mathcal{G}(\tau_{k+1}, \tau_k)$ for $j > k$, and $\mathcal{G}(\tau_k, \tau_k) = \mathbf{I}$. Repeated application of the

recurrence yields:

$$e_N = \mathcal{G}(\tau_N, \tau_0)e_0 + \sum_{k=0}^{N-1} \mathcal{G}(\tau_N, \tau_{k+1}) \cdot \left(\int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h})) - h^*(\mathbf{x}^*(u), u) \right] du + \delta_k \right). \quad (\text{A.14})$$

Substituting $\tau_N = 0$, $\tau_0 = T$, $e_N = \hat{\mathbf{x}}_N - \mathbf{x}^*(0)$, and $e_0 = \hat{\mathbf{x}}_0 - \mathbf{x}^*(T)$:

$$\hat{\mathbf{x}}_N - \mathbf{x}^*(0) = \underbrace{\mathcal{G}(0, T)(\hat{\mathbf{x}}_0 - \mathbf{x}^*(T))}_{\text{propagated initial error}} + \sum_{k=0}^{N-1} \mathcal{G}(0, \tau_{k+1}) \cdot \left(\underbrace{\int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) - h^*(\mathbf{x}^*(u), u) \right] du}_{\text{single step model prediction error contribution}} + \underbrace{\delta_k}_{\text{single step truncation error}} \right). \quad (\text{A.15})$$

where the propagated initial-error term represents the deterministic contribution of a particular initialization error to the final path error, with its statistical origin in the distributional Diffused-Prior Gap.

A.3 THE DISCREPANCY BETWEEN DDPM AND DDIM IN THE FORWARD AND REVERSE PROCESS

As shown in Eq. (2), the DDPM (Ho et al., 2020) forward process can be written as:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + (1 - \bar{\alpha}_t) \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (\text{A.16})$$

The corresponding reverse process is defined as:

$$\mathbf{x}_{t-1}^{\text{DDPM}} = \frac{1}{\sqrt{\bar{\alpha}_t}} \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\bar{\alpha}_t} \sqrt{1 - \bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) + \sqrt{\beta_t} \boldsymbol{\epsilon}. \quad (\text{A.17})$$

For DDIM (Song et al., 2021), the forward process is identical to Eq. (A.16), whereas the deterministic reverse process with $\eta = 0$ is:

$$\begin{aligned} \mathbf{x}_{t-1}^{\text{DDIM}} &= \sqrt{\frac{\bar{\alpha}_{t-1}}{\bar{\alpha}_t}} (\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)) + \sqrt{1 - \bar{\alpha}_{t-1}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \\ &= \frac{1}{\sqrt{\bar{\alpha}_t}} \mathbf{x}_t - \left(\frac{\sqrt{1 - \bar{\alpha}_t}}{\sqrt{\bar{\alpha}_t}} - \sqrt{1 - \bar{\alpha}_{t-1}} \right) \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t). \end{aligned} \quad (\text{A.18})$$

Although DDPM and DDIM share the same forward diffusion process, DDIM is not obtained merely by removing the stochastic term from the DDPM reverse denoising process. Rather, the two methods use different coefficient formulations for the predicted noise term.

A.4 DATASET EXAMPLES AND DETAILS

Figure A1 presents representative examples from the three datasets used in this study. As shown in Table A1, each dataset is balanced across its respective categories; Table A2 further shows that images with different object counts are comparably distributed, indicating the absence of object-count bias across datasets. The datasets follow the same ordering in both tables, and the category order in Table A1 is defined as follows:

- **ToyShape**: triangle, square, pentagon;
- **SimObject**: mug, apple, clock;
- **RealHand**: finger.

Table A1: Per-category object counts for each dataset.

	ToyShape	SimObject	RealHand
Category 1	20,027	19,916	25,250
Category 2	19,902	20,101	-
Category 3	19,978	19,974	-

Table A2: Number of images grouped by object count for each dataset.

	ToyShape	SimObject	RealHand
1 object	10,003	10,022	-
2 objects	10,087	9,965	-
3 objects	9,910	10,013	-
5 objects	-	-	25,250
Total	30,000	30,000	25,250

A.5 EXAMPLES OF COUNTING HALLUCINATIONS

We provide representative counting hallucinations for all three datasets, as shown in Fig. A2. For ToyShape and SimObject, counting hallucinations either contain more than two instances of at least one category or contain no object instances despite exhibiting a normal background, as shown in Fig. A2(a) and Fig. A2(b). For RealHand, counting-hallucinated samples are characterized by incorrect valid-finger counts, such as 4 or 6 fingers, as shown in Fig. A2(c). Notably, the final two images in the first row of Fig. A2(c) appear at first glance to contain five fingers; however, one finger is partially incomplete and visibly artifacted, effectively leaving four valid fingers. Accordingly, these samples are classified as counting hallucinations.

A.6 EXAMPLES OF YOLO PREDICTIONS OF FINGERTIPS

Figure A3 presents representative YOLO predictions on generated RealHand images. The detector accurately localizes fingertips across diverse hand poses and orientations, including both dorsal and palmar views. Nevertheless, it may fail to distinguish anatomically implausible but count-valid configurations, such as hands with duplicated thumbs (e.g., the penultimate sub-figure of the first row and the last sub-figure of the last row in Fig. A3(a)) or hands with five middle fingers but no thumb (e.g., the last sub-figure of the first row and the penultimate sub-figure of the last row in Fig. A3(a)). These anatomically abnormal yet counting-valid samples are still detected as five-finger hands and therefore remain indistinguishable from anatomically valid samples under a purely count-based criterion.

Figure A3(b) shows counting-hallucinated samples for which YOLO predicts incorrect fingertip counts, such as 3, 4, 6, or 7 fingertips. Figure A3(c) further illustrates rare edge cases (<1%) in which the detector fails to identify fingertips with unclear boundaries, cracked fingers, or floating artifacts. Although such cases can be partially mitigated by tuning YOLO inference hyperparameters, including IoU and confidence thresholds, doing so may degrade detection performance for the majority of counting-valid samples. Because all RealHand evaluations use a consistent detection setup, these rare cases do not alter the validity of the overall conclusions, even though their complete resolution remains difficult and offers limited practical reward.

A.7 EXAMPLES OF THREE CLASSES OF GENERATED HAND IMAGES

Figure A4 presents representative examples from the three generated-hand categories: counting-valid samples, counting-hallucinated samples, and non-counting failures. As illustrated in Fig. 2, the RealHand evaluation pipeline first applies the counting-ready indicator to separate countable samples, including counting-valid and counting-hallucinated images, from images unsuitable for counting. The latter typically contain severe visual artifacts, including distorted, cracked, partially visible, or

floating fingers. Counting-ready samples are then classified as counting-valid or counting-hallucinated according to whether the detected number of valid fingertips equals five.

A.8 EXTENDED DETAILS OF PTI-HAND EXPERIMENTS

In the PTI-Hand experiments, we use 10 distinct hand-generation prompts with Stable-Diffusion-3.5-Medium. The prompts are listed below.

1. A single human hand, front view, all fingers in frame and visible.
2. A complete human hand, seen from the front, showing all fingers clearly, minimal background.
3. An isolated single hand, viewed from directly above, showing all fingers, simple studio background.
4. Overhead shot of a person’s hand, entire hand in frame, all fingers visible, plain background.
5. One hand, top view, five fingers visible, simple background.
6. A single hand, front or back view, flat against a surface, all fingers in frame and fully visible.
7. A single human hand, top-down view, all fingers visible, resting on a plain white surface.
8. A single human hand seen from above, fingers spread naturally, minimal background.
9. A human hand, top view, palm facing downward, all fingers clearly in frame, neutral background.
10. A single human hand captured from overhead, fingers extended, plain background with soft lighting.

The hallucination definition presupposes that the model generates configurations absent from its training data. For large-scale pretrained diffusion models, however, this presupposition cannot be directly verified because the full training distribution is inaccessible. To make counting-hallucination quantification feasible in this setting, we adopt a conservative criterion: any generated hand image containing six or more fingers is treated as counting-hallucinated. This criterion is well justified because, although real images may show fewer than five visible fingers because of occlusion, articulation, or viewpoint, real data are highly unlikely to contain hands with six or more fingers. Thus, finger counts ≥ 6 provide a reliable and empirically verifiable signal of counting hallucination; generated examples are shown in Fig. A5.

A.9 EXTENDED DETAILS, RESULTS, AND DISCUSSION OF THE JOINT DIFFUSION MODEL

A.9.1 IMPLEMENTATION DETAILS IN TABLE 3

In the JDM experiments reported in Table 3, we fine-tune the denoising UNet for 120k steps, compared with 80k steps for the vanilla LDM. The larger training budget is necessary because expanding the latent space from three to six channels in JDM creates a substantial mismatch with the pretrained latent distribution of the LDM. Additional optimization is therefore required to reduce this distributional gap and allow JDM’s modeling capacity to materialize fully.

A.9.2 FINE-TUNING PRETRAINED TEXT-TO-IMAGE MODELS WITH JOINT-DIFFUSION TRAINING

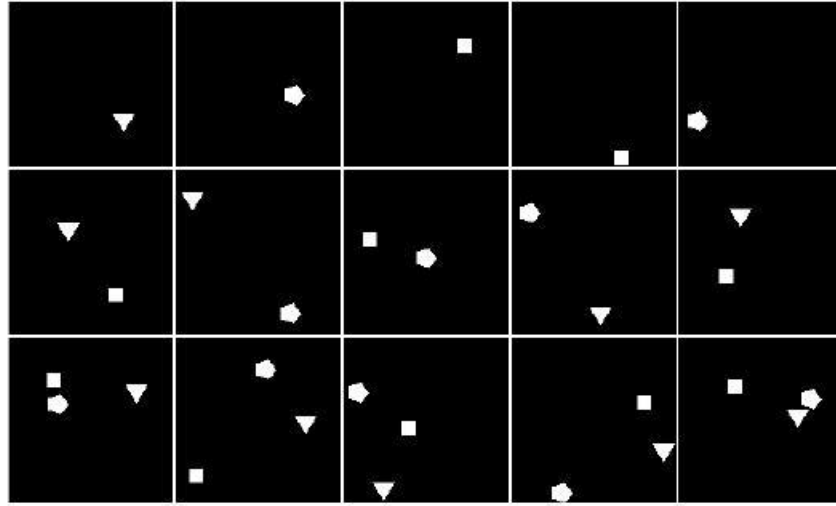
Importantly, JDM should be interpreted as a **novel training paradigm** for improving factual generation in diffusion models rather than as a superior inference strategy. On the one hand, JDM extends only the input and output channels of the UNet while preserving the sizes of all intermediate layers, thereby **introducing negligible inference-time overhead relative to vanilla LDM**. On the other hand, augmenting a pretrained text-to-image model with additional mask channels **disrupts the manifold geometry of the pretrained latent space and alters the original denoising distribution**, making direct fine-tuning of such a model nontrivial.

Table A3: Comparison of failure rates and image quality between RealHand fine-tuning of SD-1.5 and RealHand fine-tuning of SD-1.5 under the joint-diffusion training paradigm. The first row reports the baseline obtained by fine-tuning Stable-Diffusion-1.5 on RealHand data. The second row reports results obtained by freezing all intermediate layers of the baseline model and fine-tuning only the input and output layers with extended channel dimensions. Lower values indicate better performance.

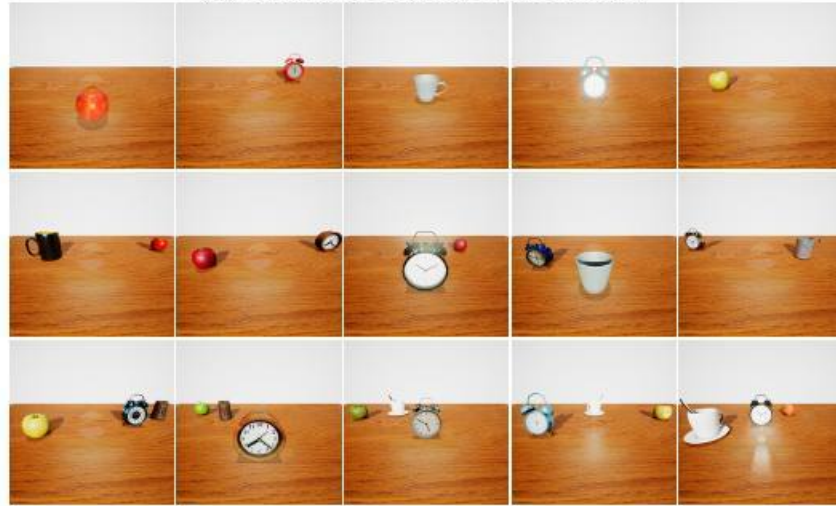
Solver Name	Counting Hallucination Rate (CHR) ↓						Non-counting Failure Rate (NCFR) ↓						Fréchet Inception Distance (FID) ↓					
	DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2		
Sampling Steps	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100
SD-1.5	0.3362	0.3404	0.3630	0.3396	0.3376	0.3362	0.0059	0.0032	0.0020	0.0050	0.0026	0.0018	61.51	57.71	57.70	58.53	57.57	56.69
SD-1.5 with joint-diffusion finetuning	0.2838	0.3222	0.4145	0.3053	0.3277	0.3558	0.0947	0.0511	0.0487	0.0549	0.0362	0.0327	70.65	79.22	102.39	70.80	72.84	83.07

Despite this challenge, we observe promising reductions in counting hallucinations when JDM is applied to large-scale pretrained diffusion models under carefully controlled fine-tuning conditions. In practice, jointly aligning the extended JDM latent space with the highly rigid latent geometry of large pretrained models is difficult at the limited scale of RealHand. To balance feasibility and adaptation, we use a two-stage fine-tuning strategy: first, we fine-tune Stable-Diffusion-1.5 on RealHand; then, we freeze all intermediate UNet layers and fine-tune only the input and output layers with extended channels. As shown in Table A3, even under this highly constrained fine-tuning setup, the joint-diffusion paradigm consistently yields lower counting hallucination rates at 25 and 50 sampling steps across both solvers.

Overall, the evidence presented here suggests that JDM is a promising direction for enforcing factual consistency in image generation. Finally, **exploring its behavior and performance under genuinely large-scale training remains an important and compelling direction for future research.**



(a) Representative ToyShape dataset examples

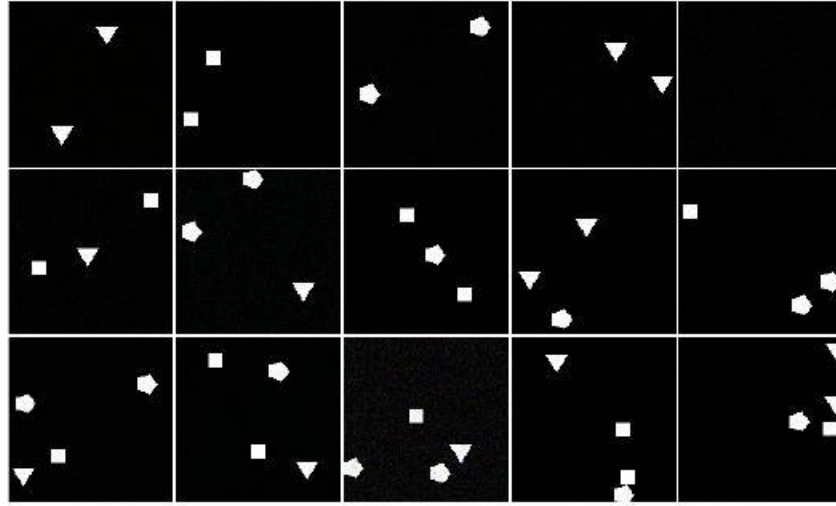


(b) Representative SimObject dataset examples

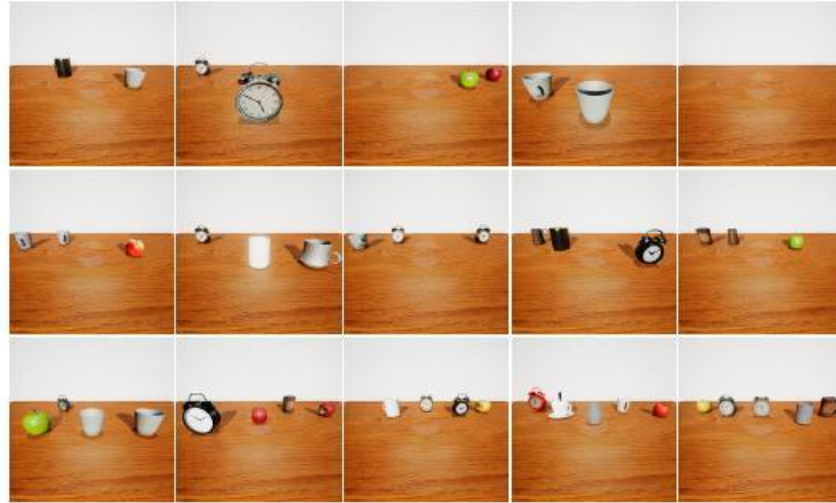


(c) Representative RealHand dataset examples

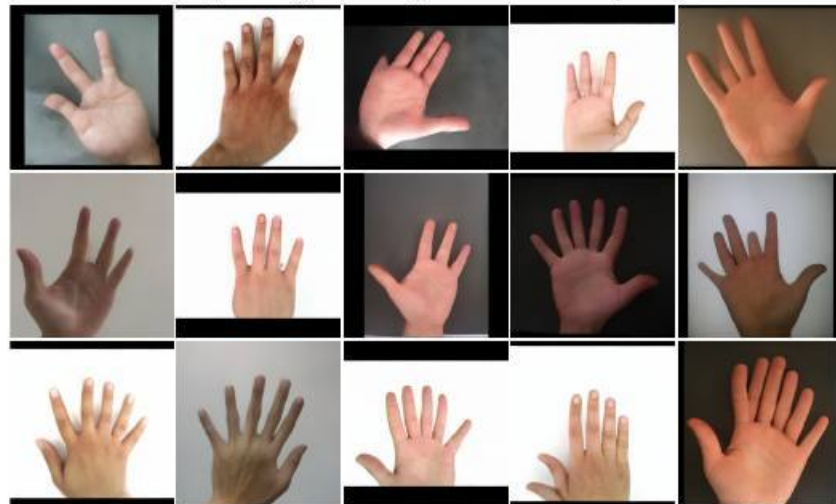
Figure A1: Representative examples from the CountHalluSet suite, comprising ToyShape, SimObject, and RealHand.



(a) ToyShape counting-hallucination examples

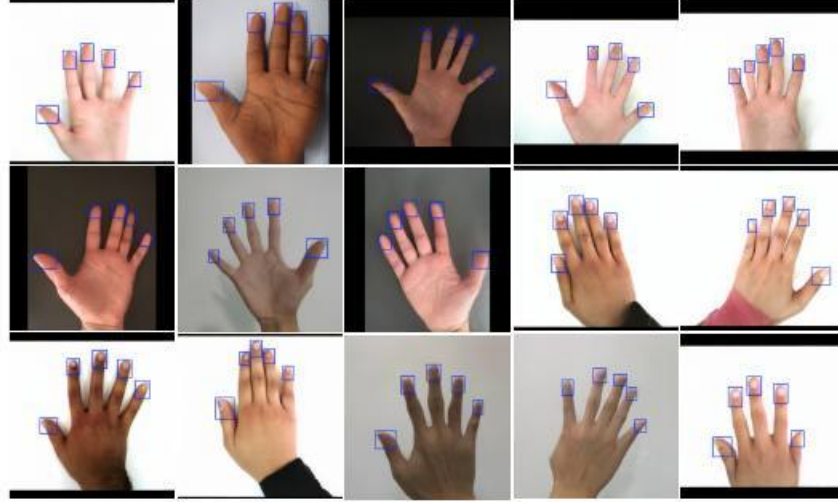


(b) SimObject counting-hallucination examples

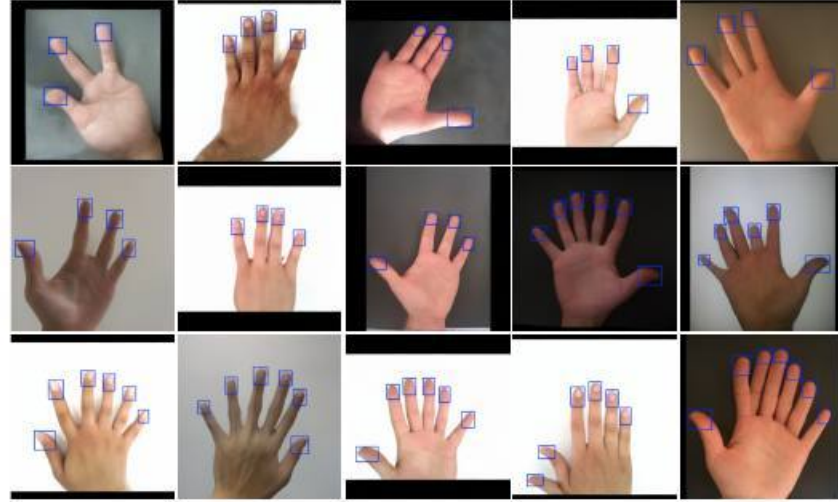


(c) RealHand counting-hallucination examples

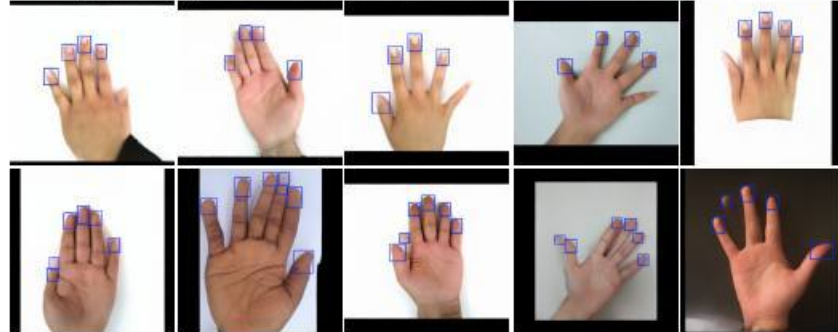
Figure A2: Representative counting-hallucination examples for ToyShape, SimObject, and RealHand.



(a) Counting-valid YOLO predictions with five detected fingers



(b) Counting-hallucinated YOLO predictions with non-five-finger counts



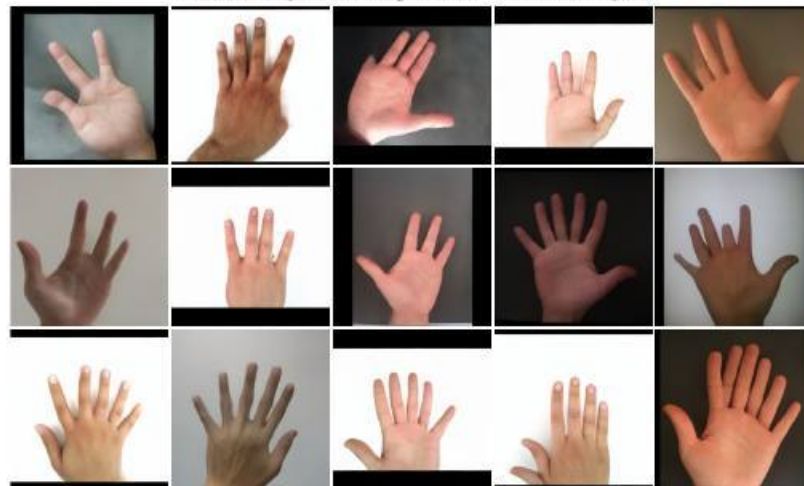
(c) Rare edge cases in which YOLO predictions fail

Figure A3: YOLO prediction examples on generated RealHand images. (a) Counting-valid samples for which YOLO predicts five fingertip bounding boxes, including cases with two thumbs. (b) Counting-hallucinated samples with incorrect fingertip-count predictions, such as 3, 4, 6, or 7 detected fingertips. (c) Rare edge cases that are not filtered by the counting-ready indicator and remain difficult for YOLO, including blurred, incomplete, or anatomically disconnected fingers.

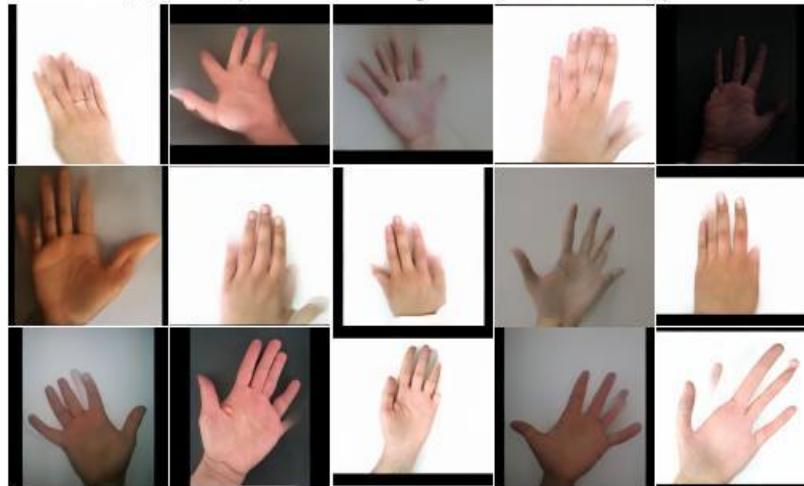
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(a) Counting-valid samples with five valid fingers



(b) Counting-hallucinated samples without five valid fingers



(c) Non-counting failures unsuitable for counting

Figure A4: Examples of the three generated-hand classes. (a) Counting-valid images contain five clearly discernible fingers. (b) Counting-hallucinated images contain an incorrect number of valid fingers, such as 4 or 6. (c) Non-counting failures are unsuitable for counting because of severe visual artifacts, including distorted, cracked, partially visible, or floating fingers.



(a) Counting-valid samples with five valid fingers



(b) Counting-hallucinated samples with more than 5 fingers



(c) Non-counting failures unsuitable for counting

Figure A5: Examples of the three generated-hand classes produced by pretrained Stable-Diffusion-3.5-Medium (PTI-Hand). (a) Counting-valid images contain five clearly discernible fingers. (b) Counting-hallucinated images contain more than five valid fingers, such as 6 or 7. (c) Non-counting failures are unsuitable for counting because of severe visual artifacts, including distorted, cracked, partially visible, or floating fingers.